

1

Introduction

It has been nearly twenty years since the issue of digital assets and death was first discussed in the media. Nevertheless, there is still little clarity in the law in most places. In 2005, a judgment was handed down in the US, *In re Ellsworth*,¹ where the family of a US Marine, Justin Ellsworth, killed in Iraq, requested access to his emails stored in his Yahoo! account. Yahoo! refused, referring to its terms of service, which banned access to all third parties on death. Yahoo! claimed that the law prohibits it from disclosing deceased users' personal communications without a court order. The family argued that as his heirs, they should be given access to Justin's emails and the account, his sent and received emails, as his last words. Yahoo!, on the other hand, had a non-survivorship policy, and there was a risk that Ellsworth's account could have been deleted for good. The judge allowed Yahoo! to enforce its privacy policy and did not order the transfer of the account credentials. Instead, he made an order requiring Yahoo! to enable access to the deceased's account content by providing the family with a CD containing copies of the emails. The media reported that Yahoo! initially provided the emails received by Justin Ellsworth on a CD and, after the family complained again, allegedly subsequently sent paper copies of the emails.²

In a more recent case in Germany, the parents of a fifteen-year-old girl who allegedly committed suicide wanted to access the account and private messages to understand why it happened. Facebook refused to grant access to the deceased girl's Facebook account to her parents, arguing that its duty

¹ No 2005-296, 651-DE (Mich Prob Ct 2005).

² A Kulesza, 'What happens to your Facebook account when you die?' *Blog* (3 February 2012) <<http://blogs.lawyers.com/2012/02/what-happens-to-facebook-account-when-you-die/>>; Associated Press release, *justinellsworth.net* (21 April 2005) <<http://www.justinellsworth.net/email/ap-apr05.htm>> last accessed 30 January 2022.

of confidentiality was owed to her and her contacts.³ The Court of Appeal in Berlin upheld Facebook's decision, but the Federal Court of Justice overturned this, applying the law and the principle of universal succession (i.e. that heirs 'step into the deceased's shoes' for all her rights and obligations).⁴

In another US case, *In re Scandalios*,⁵ Ric Swezey died unexpectedly in 2017. His will did not explicitly authorise his husband, Nicholas Scandalios, to access his digital assets, including many family photos in his Apple account. Unsurprisingly, Apple iCloud's terms and conditions stated that 'any rights to your Apple ID or content within your account terminate upon your death', unless required by law. However, the New York County Surrogate's Court ordered Apple to give the deceased's husband and the executor of the estate access to the deceased's Apple account. In 2015, Rachel Thompson was denied access to her late husband's phone in a similar UK case, leaving the family photos and videos locked away. Rachel wanted to gain access to her husband's account and share the photos and videos with their daughter. Apple refused this request and said they would only grant her access if she got a court order to that effect. A court order was made in 2019 by the Central London County Court judge.⁶ The judge commented that the law should be reformed to provide clarity in future cases, which are likely to be frequent.

We have seen numerous other court cases and media stories that resemble those above. More recently, Western media have reported an increased number of stories involving the creation of deep fakes and chatbots of the deceased, which raised a few eyebrows on Twitter and sparked patchy interest in this area. It also reminded some of the Black Mirror episode 'Be Right Back'⁷ and all the ethical, legal and social issues it illustrated so aptly.⁸

³ BBC News, 'Facebook ruling: German court grants parents rights to dead daughter's account' *BBC News* (12 July 2018) <<https://www.bbc.com/news/world-europe-44804599>> last accessed 30 January 2022.

⁴ Kammergericht, Urteil vom 31. Mai 2017, Aktenzeichen 21 U 9/16 <<https://www.berlin.de/gerichte/presse/pressemitteilungen-der-ordentlichen-gerichtsbarkeit/2017/pressemitteilung.596076.php>> last accessed 30 January 2022; appealed BGH, 12 July 2018, Docket No III ZR 183/17.

⁵ 2017-2976/A NY Surr Ct 2019.

⁶ Mark Bridge, 'Widow wins long battle for iPhone family photos' *Times* (19 May 2019) <<https://www.thetimes.co.uk/article/widow-wins-long-battle-for-iphone-family-photos-h7mv9bw7t>> last accessed 30 January 2022.

⁷ Season 2, Episode 1, 11 February 2013. Series currently available on Netflix and More4.

⁸ For a commentary, see L Edwards and E Harbinja, "'Be right back": What rights do we have over post-mortem avatars of ourselves?' in L Edwards, B Schafer and E Harbinja (eds), *Future Law: Emerging Technology, Regulation and Ethics* (Edinburgh University Press 2020).

This book tackles key legal and ethical issues illustrated in the scenarios above and elsewhere. It provides the first comprehensive academic account of the transmission of digital assets on death in the UK and more widely.⁹ The book also builds on the most relevant research around digital death and dying in the humanities and social sciences.¹⁰

To achieve this overall aim, in Chapters 2 and 3, I will set out the book's theoretical framework. In Chapter 2, I will consider the theory and doctrine of property, and in Chapter 3, the focus will shift to considerations of autonomy, testamentary freedom and post-mortem privacy. This will enable me to set out an adequate background and underpinnings for case studies in Chapters 4, 5 and 6 and offer suggestions and solutions in Chapter 7.

The first part of the book is thus primarily theoretical, looking at theories and doctrines of property, autonomy, post-mortem privacy, postmortal privacy and digital immortality, which are the key questions to answer before looking at digital legacy and the legal nature of digital assets in more detail.¹¹ The consideration of property is vital for one fundamental reason. If a digital asset is property, this has to be established clearly at the outset. If an asset can be considered property of the deceased user, it forms a part of an estate and transmits on death in most countries. The same is valid for intellectual property (IP) (primarily copyright), arguably a subset of property. Even if we reject the categorisation of IP as property, copyright protection transmits on death and lasts for seventy years post-mortem in most Western countries. Whereas copyright protection has been harmonised to a great extent within these jurisdictions, the doctrinal and normative conceptions of property vary by jurisdiction, especially regarding common and civil law. This lack of harmonisation is vital since digital assets are typically located in a transnational space, for example on a server physically located in Ireland, owned

⁹ E Harbinja, *Legal Aspects of Transmission of Digital Assets on Death* (PhD thesis, University of Strathclyde 2017).

¹⁰ MH Jacobsen (ed.), *Postmortal Society: Towards a Sociology of Immortality* (Routledge 2017); R Kastenbaum, *On Our Way: The Final Passage through Life and Death (Life Passages)* (University of California Press 2004); C Öhman and L Floridi, 'The political economy of death in the age of information: A critical approach to the digital afterlife industry' (2017) *27 Minds & Machines* 639.

¹¹ In addition to the above literature on death and dying, the theory part focuses on key justifications of property and incidents of property in the works of Bentham, Locke, Hegel, Radin, Honoré, Hohfeld, Penner, Bridge and Marx. In terms of autonomy and privacy, key authors that the book considers are Mill, Locke, Bentham, Rawls, Hegel, Henkin, Cohen, Nissenbaum, Schwartz, Samuelson, Desai, Edwards, Schafer, Floridi and Bernal. In terms of post-mortem and postmortal privacy, I build on my own work in the area and the work of Birnhack, Desai, Edwards, Buitelaar and Malgieri.

by a US company, and accessed by and created by users from many different jurisdictions.

Conversely, if a digital asset is not an object of property, it would be an aspect of post-mortem privacy, which I started developing earlier in my work, referring to theories of autonomy, privacy and data protection.¹² This concept is then used as a basis for a new concept of postmortal privacy, which includes philosophical conceptions of the informational body, sociological research around digital immortality and technological developments of artificial intelligence (AI).¹³

I will then test these theories and issues of contract law and copyright on chosen case studies of digital assets. Digital assets include a wide variety of different assets, and their number is growing with the development of new technologies (business accounts, emails, social networks, games, personal data, domain names, virtual currencies, etc.). Due to their number and features, it is argued that it would not be viable to look at the legal nature, and consequently the transmission on death, of all these different assets. Therefore, using the theoretical framework set out in the following two chapters, the book analyses the three most typical and widely used types of assets, that is, emails, social networks and virtual worlds and games (virtual realities). It is not argued here that these assets are the most economically valuable or will continue being as significant as technology develops. Instead, the examples are chosen for their current prominence, usage, user base and complexities surrounding their legal nature. These assets are perhaps most intrinsically intertwined in the everyday life of an average user and, since the focus of the book is on the users and not on the businesses, the examples are appropriate. I have already analysed aspects of the transmission of the chosen digital assets in my earlier work.¹⁴ This book will develop this research further, update and consolidate findings and recommendations, and offer novel theoretical and practical considerations under a more coherent legal framework.

¹² In particular, E Harbinja, 'Post-mortem privacy 2.0: Theory, law and technology' (2017) 31 *International Review of Law, Computers & Technology* 26; Harbinja, *Legal Aspects* (n 9).

¹³ E Harbinja, 'The "new(ish)" property, informational bodies, and postmortality' in M Savin-Baden and V Mason-Robbie (eds), *Digital Afterlife: Death Matters in a Digital Age* (Taylor & Francis 2020) 89–106.

¹⁴ E Harbinja, 'Virtual worlds – a legal post-mortem account' (2014) 11:3 *SCRIPTed* 273; E Harbinja, 'Virtual worlds players – consumers or citizens?' (2014) 3(4) *Internet Policy Review* 1; E Harbinja, 'Legal nature of emails: A comparative perspective' (2016) 14 *Duke Law and Technology Review* 227; E Harbinja, 'Social media and death' in L Gillies and D Mangan (eds), *The Legal Challenges of Social Media* (Edward Elgar 2017); E Harbinja, 'Emails and death: Legal issues surrounding post-mortem transmission of emails' (2019) 43(7) *Death Studies* 435.

The book will offer theoretical arguments around post-mortem and post-mortem privacy, suggestions for law reforms, technology solutions and developments, and practical advice for users. These arguments and the framework can further be used to test new types of digital assets, their legal nature, and their transmission on death. Criminal law and jurisdiction will be discussed but are not a primary consideration for the book, as these areas deserve a separate in-depth discussion and analysis.

1.1 The Concept and Value of Digital Assets

The notion of digital assets is a relatively new phenomenon, lacking a proper legal definition, with diverse meanings attributed to it. From a lay person's perspective, digital assets could be anything valuable online, any asset (account, file, document, digital footprint) that has a personal, economic or social attachment to an individual. The legal meaning, however, needs a little more precision. Determining the legal definition and nature is essential to enable adequate legal treatment and regulation. So far, there have been a few attempts to define and classify digital assets. Most definitions are inductive and try to theorise starting from the existing assets online, trying to make appropriate generalisations and classifications. Many authors interchangeably use the terms 'virtual assets' and 'digital assets'. In this book, for precision and consistency, the term 'digital assets' will be used as an umbrella term unless otherwise stated. The term 'virtual assets' will be reserved for Chapter 5 and considerations of assets in virtual worlds.

One of the most comprehensive definitions so far has been offered by Cahn. She categorises digital assets into the following: personal assets ('typically stored on a computer or smartphone or uploaded onto a website, including photographs, videos, or even music playlists'¹⁵), social media assets ('including websites such as Facebook and Twitter, as well as email accounts'¹⁶), financial assets ('bank accounts, Amazon accounts, Pay-Pal accounts, accounts with other shopping sites, or online bill payment systems'¹⁷, and virtual currency) and business accounts (which 'generally include customer addresses and patient information'¹⁸). Perrone accepts and uses this categorisation.¹⁹

¹⁵ N Cahn, 'Postmortem life on-line' (2011) 25 *Probate & Property* 36, 36–7.

¹⁶ Ibid.

¹⁷ Ibid.

¹⁸ Ibid.

¹⁹ M Perrone, 'What happens when we die: Estate planning of digital assets' (2012/13) 21 *CommLaw Conspectus* 185.

Due to their unique features, some assets could be included in all these categories. One example is an eBay or Instagram reputation or following, which is personal, dependent on the social interactions (user's feedback), tied to a business account and brings financial benefits. With the increased integration of online services, many platforms now include all of these categories simultaneously. For instance, Gmail, Google's email platform, might be used for business purposes, for storing photographs and videos (connected to Google Drive, Google's cloud storage service), for social network purposes (as it is connected to Google+, Google's social network site) and as a payment system (with the recent feature of sending money via Gmail, though a user's Google Wallet service in the US and UK). It is, therefore, sometimes difficult to separate and define the categories of digital assets.

Another categorisation places digital assets into the following categories: access information, tangible digital assets, intangible digital assets and metadata.²⁰ The access information includes account numbers and login information, which, according to Haworth, are not assets in the strict sense as they only enable access to other assets. On the other hand, tangible digital assets are digital property, held in a definable form, which are likely to be transferred and converted into physical assets (photos, documents, emails, online banking account balances, domain names, blog posts).²¹ Further, intangible digital assets are those harder to conceptualise, spread over the Internet in volumes and likely needing to be deleted or shut down ('likes' on Facebook, website profiles, comments, reviews). Lastly, according to Haworth, metadata ('data about the data', histories, deleted data, code, location tags, etc.) encounters similar issues to intangible assets, being even harder to find and gain access to.²² Therefore, in her opinion, only the category of tangible digital assets is assets and digital property *stricto sensu*.

This categorisation is quite problematic. First, most of the assets categorised as tangible for this definition will never really be converted to a physical, offline form. Log files and metadata can hardly be conceived of as digital assets, as these are just properties of an underlying system and not something that has an individual and independent value and existence. Metadata signifies some properties of other data and is derived from them, and login data serves as a tool provided by service providers for users to gain access to their other assets. As a side note, metadata can be valuable to service providers, as it provides critical analytics and indicates users' behaviours. Therefore,

²⁰ S Haworth, 'Laying your online self to rest: Evaluating the Uniform Fiduciary Access to Digital Assets Act' (2014) 68 *University of Miami Law Review* 535, 538.

²¹ Ibid.

²² Ibid. 539.

Haworth's definition is problematic, just like those of Hopkins and Băbeanu et al., who argue that metadata, a valuable piece of data, can also represent a type of digital asset and help detect and find other digital assets.²³

A more general definition starts with defining the terms that coin the notion of digital assets. For instance, the *Oxford English Dictionary* defines digital as 'Of signals, information, or data: represented by a series of discrete values (commonly the numbers 0 and 1), typically for electronic storage or processing.'²⁴ Similarly, 'virtual' is defined as 'occurring or existing primarily online' or 'being simulated on a computer or computer network'. According to this definition, 'virtual assets are the electronic information stored on a computer or through computer-related technology'.²⁵ Hopkins defines a digital asset as an asset that 'exists only as a numeric encoding expressed in binary form' or 'any electronically stored information'.²⁶ Importantly, as rightly noted by Haworth, any definition of digital assets needs to be both broad (to encompass innovations online) and still clear enough so that everyone understands what it means.²⁷

The US Uniform Law Commission Fiduciary Access to Digital Assets Committee proposed an all-encompassing definition of digital property in its first draft,²⁸ which includes both digital assets and digital accounts (providing

²³ JP Hopkins, 'Aferlife in the cloud: Managing a digital estate' (2013) 5 *Hastings Science and Technology Law Journal* 210, 211; D Băbeanu et al., 'Strategic outlines: Between value and digital assets management' (2009) 11 *Annales Universitatis Apulensis Series Oeconomica* 318, 319.

²⁴ *Oxford English Dictionary* (OED) (online edn, Oxford University Press 2016) <<http://www.oed.com/>> last accessed 1 August 2021.

²⁵ American Law Institute, American Bar Association Continuing Legal Education 'Representing estate and trust beneficiaries and fiduciaries: Virtual assets' *ALI-ABA Course of Study* (14–15 July 2011) 1 <http://www.cobar.org/repository/Inside_Bar/TrustEstate/SRC/Virtual%20Asset%20Subcommittee%20Research%20%231.pdf> last accessed 1 August 2021.

²⁶ Hopkins (n 23) 202. See similarly N Dosch, 'Overview of digital assets: Defining digital assets for the legal community' *Digital Estate Planning* (14 May 2010) <<http://www.digitalestateplanning.com/>> last accessed 1 August 2021, who defines a digital asset as 'any file on your computer in a storage drive or website and any online account or membership'. Conner accepts and uses this definition. See J Conner, 'Digital life after death: The issue of planning for a person's digital assets after death' (2010–11) 3 *Estate Planning & Community Property Law Journal* 301, 303.

²⁷ Haworth (n 20) 3.

²⁸ National Conference of Commissioners on Uniform State Laws, Drafting Committee on Fiduciary Access to Digital Assets, 'Fiduciary Access to Digital Assets Act' (15–16 February 2013 Drafting Committee Meeting) <http://www.uniformlaws.org/shared/docs/Fiduciary%20Access%20to%20Digital%20Assets/2013feb7_FADA_MtgDraft_Styled.pdf> last accessed 1 August 2021.

access to a digital asset or a digital service).²⁹ The second draft, from May 2013, retains this definition, clarifying that digital property does not include the content of electronic communication. However, the October 2013 draft seems less ambitious in its definition, discarding the notion of digital property and retaining only the revised concept of digital assets.³⁰ The July 2014 draft of the Act revised the definition once more, namely:

‘Digital assets’ include products currently in existence and yet to be invented that are available only electronically. Digital assets include electronically stored information, such as: 1) any information stored on a computer and other digital devices; 2) content uploaded onto websites, ranging from photos to documents; and 3) rights in digital property, such as domain names or digital entitlements associated with online games. . . . Both the catalogue and content of an electronic communication are covered by the term ‘digital assets’.³¹

Therefore, the definition is quite inclusive and technologically neutral, as it leaves room for assets ‘yet to be invented’. In addition, it includes different general types of content, such as information, content uploaded online, rights and catalogues of electronic communications (meaning log files). The definition, however, expressly excludes ‘an underlying asset or liability unless the asset or liability is itself an electronic record’.³² The Uniform Law Commission (ULC) revised the Act in December 2015, narrowing down the definition further, including only electronic records ‘in which an individual has a right or interest’. Similarly to the previous draft, the ULC expressly excludes underlying assets or liabilities.³³

²⁹ Ibid. ‘(8) “Digital asset” means information created, generated, sent, communicated, received, or stored by electronic means on a digital service or digital device and includes, without limitation, any usernames, words, characters, codes, or contract rights pursuant to the terms of service agreement that controls access to a digital account.’

³⁰ National Conference of Commissioners on Uniform State Laws, Drafting Committee on Fiduciary Access to Digital Assets, ‘Fiduciary Access to Digital Assets Act’ (22 October 2013) <http://www.uniformlaws.org/shared/docs/Fiduciary%20Access%20to%20Digital%20Assets/2013nov_FADA_Mtg_Draft.pdf> last accessed 1 August 2021.

³¹ National Conference of Commissioners on Uniform State Laws, Drafting Committee on Fiduciary Access to Digital Assets, ‘Fiduciary Access to Digital Assets Act’ (July 2014), s 2(8) and p 9 <http://www.uniformlaws.org/shared/docs/Fiduciary%20Access%20to%20Digital%20Assets/2014_UFADAA_Final.pdf> last accessed 1 August 2021.

³² Ibid. s 2(8).

³³ National Conference of Commissioners on Uniform State Laws, Drafting Committee on Fiduciary Access to Digital Assets, ‘Revised Fiduciary Access to Digital Assets Act’ (December 2015), s 2(10) <[http://www.uniformlaws.org/Act.aspx?title=Fiduciary%20Access%20to%20Digital%20Assets%20Act,%20Revised%20\(2015\)](http://www.uniformlaws.org/Act.aspx?title=Fiduciary%20Access%20to%20Digital%20Assets%20Act,%20Revised%20(2015))> last accessed 1 August 2021.

A similar definition has been crafted by Lamm, a US probate attorney, emphasising that the concept includes IP and contractual rights.³⁴ Lamm also notes that the full access to a standard account ‘isn’t all that valuable to family members or fiduciaries’. He argues that ‘the contents of the online account are where the financial or sentimental value is located’.³⁵ Conner, conversely, confuses these concepts, claiming that virtual property and digital assets are synonymous.³⁶ As we shall see later in this chapter, virtual property is usually used to describe a player’s property in virtual worlds and games. In discussing property, Conner finds it essential to place digital assets in one of the traditional types of property, tangible or intangible. Thus, he concludes that it is difficult to make a clear distinction here, as digital assets could change their quality and become tangible from their initial intangible state (e.g. printing of photos).³⁷

Edwards and Harbinja define digital assets ‘widely and not exclusively to include a vast range of intangible information goods associated with the online or “digital world”’, giving examples of different digital assets.³⁸

To conclude, a digital asset is any asset of personal or economic value available only in an electronic form (capable of post-mortem transmission). Such an asset could have property quality, contractual relation, IP, personality right or personal data.

This book will use the case study method and analyse the legal nature of emails, social network accounts, and virtual world assets to recognise the practical difficulties. Other valuable assets, such as financial accounts (online banking), businesses (domain names, eBay, Amazon accounts), other personal assets (MP3 collections, photograph collections, etc.) are outside the scope of this book.

³⁴ J Lamm, ‘To my son, I leave all my passwords’ *Trusts and Estate Magazine* (July 2009) <http://www.gpmlaw.com/portalresource/lookup/wosid/cp-base-4-5968/media.name=/To_My_Son_I_Leave_All_My_Passwords.pdf>; J Lamm, ‘What is digital property?’ *Digital Passing Blog* (21 June 2010) <<http://www.digitalpassing.com/2010/06/>>; J Lamm, ‘Digital property created on the Internet every 60 seconds’ *Digital Passing Blog* (20 June 2011) <<http://www.digitalpassing.com/2011/06/20/digital-property-created-internet-every-60-seconds/>> all last accessed 1 August 2021.

³⁵ J Lamm, ‘Planning ahead for access to contents of a decedent’s online accounts’ *Digital Passing Blog* (9 February 2012) <<http://www.digitalpassing.com/2012/02/09/planning-ahead-access-contents-decedent-online-accounts/>> last accessed 1 August 2021.

³⁶ Conner (n 26) 25.

³⁷ Ibid.

³⁸ L Edwards and E Harbinja, “‘What happens to my Facebook profile when I die?’: Legal issues around transmission of digital assets on death’ in C Maciel and V Pereira (eds), *Digital Legacy and Interaction: Post-Mortem Issues* (Springer 2013) 115.

We will, therefore, explore a few typical and currently relevant digital assets as examples and case studies, that is, emails, social networks sites and virtual worlds/games. As technology develops, models of protecting and transmitting these assets proposed herein could probably be more readily applicable to other kinds of emerging communications, social networks, and other virtual technologies and communities. I plan to continue exploring the nature of some other digital assets (business, financial, etc.) in future research.

The questions around the significance of this area often revolve around this critical consideration: are digital assets valuable at all? The value, of course, does not only need to be monetary; personal attachments and memories are also valuables for individuals. These are, however, harder to measure and conceptualise; their place in succession laws is not as prominent as the place of pecuniary interests since it is the primary function of this area of law to enable and facilitate the transfer of wealth.

Let us consider some general statistics first. In the UK, 92 per cent of adults and almost all adults aged sixteen to forty-four years are recent Internet users. The proportion of those aged seventy-five years and over who are recent Internet users grew from 29 per cent in 2013 to 54 per cent in 2020.³⁹ The vast majority of these users use social media, all of whom will die, leaving behind digital assets, personal data and other digital remains. Research predicts that on Facebook, for instance, the dead will outnumber the living within fifty years.⁴⁰ There is clear evidence of an increased online presence during the COVID-19 pandemic, which will result in an even greater value of digital remains. The issues are further exacerbated by the emergence of new forms of digital death industry/death entrepreneurs during the pandemic, which offer inadequate solutions for digital assets management, online mourning and grief, digital funerals, and so on (e.g. LastPass, SocialEmbers, Eternime).

So, how valuable could digital assets be? According to a 2011 survey from McAfee, Intel's security-technology unit, Americans valued their digital assets, on average, at almost \$55,000.⁴¹ World Internet users have roughly \$37,438 in digital assets across a variety of digital devices and platforms. These assets

³⁹ Office for National Statistics, 'Internet access – households and individuals, Great Britain: 2020' (7 August 2020) <<https://www.ons.gov.uk/peoplepopulationandcommunity/householdcharacteristics/homeinternetandsocialmediausage/bulletins/internetaccesshousehold-sandindividuals/2020>> last accessed 1 August 2021.

⁴⁰ C Öhman and D Watson, 'Are the dead taking over Facebook? A big data approach to the future of death online' (2019) *Big Data & Society* 1.

⁴¹ Fiduciary Access to Digital Assets Act (n 33) s 4. See also A Dale, 'More estate plans account for "digital assets"' *Wall Street Journal* (13 June 2013) <<http://online.wsj.com/article/SB10001424127887323734304578543151391292038.html>> last accessed 1 August 2021.

included: entertainment files (e.g. music downloads), personal memories (e.g. photographs), personal communications (e.g. emails, notes), personal records (e.g. health, financial, insurance), career information (e.g. resumes, portfolios, cover letters, email contacts), and hobbies and creative projects. When broken down into categories, the value shown was personal memories at around \$19,000, personal records at \$7,000, career information at \$4,000, hobbies at \$3,000, personal communications at \$3,000, and entertainment files at \$2,000. The result was an average of 2,777 digital files per person.⁴²

In the UK, in October 2011, the Centre for Creative and Social Technology (CAST) at Goldsmiths, University of London, released a study of Internet use in the UK entitled 'Generation Cloud'. The study determined that British users have at least £2.3 billion worth of digital assets stored in the cloud. The study shows that 24 per cent of UK adults estimate that they have digital assets worth more than £200 per person in the cloud, which amounts to at least £2.3bn in total.⁴³ It is also interesting to note the amount of digital content created and posted every sixty seconds online.⁴⁴ Notably, not all the assets have value and a considerable amount could be qualified as 'digital trash'.⁴⁵

The volume and value of digital legacy (often dormant) is not the only concern. Deceased individuals' personal data and aspects of their identities are being used to recreate the deceased through chatbots, holograms, deep-fakes and other similar technologies.⁴⁶ Such a recreation raises many ethical and legal problems related to autonomy, privacy and dignity. At the same time, the deceased's informational body⁴⁷ and aspects of their self and identity (digital footprints) are increasingly curated online and remain dormant, justifying an inquiry into the desirability of protection. Interestingly, some early data indicates that users express concern over post-mortem privacy

⁴² McAfee, 'McAfee reveals average Internet user has more than \$37,000 in underprotected "digital assets"' (27 September 2011) <<http://www.mcafee.com/hk/about/news/2011/q3/20110927-01.aspx>> last accessed 1 August 2021. See Hopkins (n 23) 221.

⁴³ Rackspace Hosting, 'Generation cloud: A social study into the impact of cloud-based services on everyday UK life' (16 November 2011) <http://www.rackspace.co.uk/sites/default/files/whitepapers/generation_cloud.pdf> last accessed 1 August 2021.

⁴⁴ Digital Passing, 'Digital property created on the Internet every 60 seconds' *Digital Passing Blog* (20 June 2011) <<http://www.digitalpassing.com/2011/06/20/digital-property-created-internet-every-60-seconds/>> last accessed 1 August 2021.

⁴⁵ Hopkins (n 23) 231.

⁴⁶ E Harbinja, L Edwards and M McVey, 'Governing ghostbots' (2022) *Computer Law and Security Review* (forthcoming).

⁴⁷ L Floridi, *The Fourth Revolution: How the Infosphere Is Reshaping Human Reality* (Oxford University Press 2014) 121.

when specific questions and issues are brought to their attention, but their behaviour pre-mortem does not correspond to these concerns.⁴⁸

This data concerning digital assets, further development and dominance of digital technologies and the information society, where our lives and our wealth will increasingly take on digital forms, all indicate the importance of exploring the legal nature of digital assets. Legal reality should follow the technological, economic and social one. The economic and social importance of the chosen case studies will be further elaborated in chapters that examine these assets individually.

1.2 Stakeholders

We will now identify critical stakeholders involved in any discussion around the transmission of digital assets on death. These stakeholders have different interests in digital assets, and the interests are often conflicting. Sometimes, however, they also converge, depending on the type of asset. The specific relationship between these assets will be analysed in more detail in the case study chapters. These chapters will consider the characteristics of the assets and a myriad of relationships, legal and societal, existing therein.

Users are the most significant stakeholder for this book's purposes. We start from the standpoint that users' interests are not the sole but are the paramount policy consideration in the debates surrounding digital assets. What are the reasons for this?

First, in the offline world, most legal systems already recognise that individuals can dispose of their tangible and intangible property (although this may be limited by other interests, e.g. the rights of spouses or children or society's right to inheritance taxes).

Second, the arguments for rights of testamentary freedom are arguably more robust in the online world than the offline, given the prevalence of personal data in digital assets, for example emails, social network posts, playlists, pictures, chatbots, and so on.

Third, the book draws on normative theories to question the propertisation of digital assets, one of which, personhood theory, is closely linked to the personality and creative acts of the user. This argument is powerful when looking at digital assets that also fit the category of copyright, for example online literary or musical works. If digital assets cannot be perceived as property or protected by copyright, the analysis looks at post-mortem privacy.

⁴⁸ T Morse and M Birnhack, 'The posthumous privacy paradox: Privacy preferences and behavior regarding digital remains' (2020) *New Media and Society* 1.

This concept, again, focuses on the individual, allowing for a different kind of post-mortem control that we explore in the case study chapters.

Therefore, user interests are at the centre of the argument, underpinning findings in all the chapters, that is, respecting the autonomy and wishes of the deceased, expressed pre-mortem, concerning the transmission of their assets on death.

Another group of stakeholders includes the deceased users' **heirs, next of kin and families**, on the one hand, and the **users' friends**, on the other. As shown in the hypothetical scenario earlier and in court cases analysed later in the book, the heirs and families usually aim to access the accounts/content in a digital asset, treating these as the deceased user's estate. On the other hand, friends are also interested in controlling their online relationship with the deceased, either by preserving the shared and co-constructed content or by discontinuing their relationship online. The analysis in this book does not follow the established succession law principles, where next of kin take priority in the intestate succession (the lack of a will). Instead, I recognise the shift in online and offline cultures, co-constructed and shared profiles, and the increasingly vital interests of users' friends. It is argued that the culture of sharing content online, particularly on social networks, deserves a better policy and legal recognition. Users should be able to decide and leave their assets to friends in the context of a specific digital asset, be it a social network or virtual world. A basis for this can be found in the anthropological and psychological evidence I have explored. This proposition is less applicable to the context of emails, where the feature of sharing and co-construction is not equally prominent.

Service providers and platforms (e.g. Facebook, Google, Twitter, Blizzard) have legitimate interests in preserving and developing their businesses. The providers have created their platforms, investing money and effort in them, and this should be considered. However, conventionally in the West, and especially the US, until recently, countries have tended to take a fairly *laissez-faire* attitude to the regulation of corporations. It is argued that the importance of these platforms and businesses is so significant that it merits a regulatory account. For example, Facebook's user base is enormous, more significant than any state population, and legitimate concerns of users need to be recognised (e.g. privacy concerns). In addition, the very nature of digital assets depends largely upon service providers, their computer code and servers. Borrowing from Lessig's taxonomy of regulation of the Internet,⁴⁹ it is worth noting that service providers have the power to modify, destroy and

⁴⁹ L Lessig, *Code: Version 2.0* (Basic Books 2006).

create digital assets through ‘code’, and this control cannot be left out of any considerations of digital assets. As demonstrated later in this book, solutions created by service providers support our main argument, the user’s autonomy, as they enable in-service control over a user’s content. In addition, service providers support post-mortem privacy arguments, so it is not overly difficult to consider these interests in a book focused on individual users.

Society and public interests are also worth mentioning when discussing the post-mortem treatment of digital assets. These are predominantly interests in keeping accurate historical records, archival interests, but also the potential conflict between free speech and post-mortem privacy. This perspective is not the focus of the book, however. It is argued that certain safeguards and exceptions can be established to account for these interests, but the analysis here will not discuss this in detail.

Finally, an emerging category of stakeholders is **online digital legacy services**, which aim to assist in the disposition of digital assets on death. Examples include digital wills, passwords and/or content depositories, memorial websites, messaging services and chatbots. These services aim to shift the control of digital assets to users by enabling the designation of beneficiaries who will receive the passwords/content of digital asset accounts. However, the services usually conflict with the terms of service of digital asset service providers. Furthermore, the law does not recognise them, they are not valid wills, and conflict with intestacy laws may arise. These issues will be further analysed in the final chapter, where different solutions to the issues identified in the book are assessed. However, it is worth noting here that the issues surrounding these services are significant, and it is not recommended in this book that the services are used in their current form and with the law as it stands now.

1.3 Definitions

For clarity, Table 1.1 presents definitions of the key concepts used in this book.

Table 1.1 Key Concepts Used and Developed in This Book⁵⁰

Concept	Definition	Author
Digital death	Death in the digital realm; death of an individual who leaves fragments of their identity and digital assets in the digital afterlife (e.g. digital biographies, dossiers, autobiographies and archives).	Sofka et al. 2017; Kasket 2019; Harbinja 2020
Digital immortality	Immortality reached in the digital realm as a consequence of a symbolic, a proxy or technology.	Jacobsen 2017
Digital assets	Any electronic asset of personal or economic value.	Harbinja, <i>Legal Aspects</i> 2017
	Widely and not exclusively include a huge range of intangible information goods associated with the online or 'digital world'.	Edwards and Harbinja 2013
Informational body	Inorganic body of a human constituted and existing through information related to their identity.	Floridi, 'Distributed morality' 2013; 2014
Postmortem privacy	The right of the deceased to control their personality rights and digital remains post-mortem, broadly, or the right to privacy and data protection post-mortem, narrowly defined.	Harbinja, 'Post-mortem privacy 2.0' 2017
Postmortal privacy	Protects aspects of immortality understood as the survival of the informational body, which is constituted by and exists through personal information, digital remains and memes.	Harbinja 2020

⁵⁰ Adapted from E Harbinja, 'The "new(ish)" property, informational bodies, and postmortality' in M Savin-Baden and V Mason-Robbie (eds), *Digital Afterlife: Death Matters in a Digital Age* (Taylor & Francis 2020) 90.